

STOCK PRICE PREDICTION AND FORECASTING USING MACHINE LEARNING

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ABSTRACT

Stock markets can often be one of the highly complex and most volatile places to invest. Statistical analysis of past stock performance and external factors play a major role in the decision to buy or sell stocks. Stock price index forecasting has been a subject of great research for many years, and several algorithms have been proposed to simplify this complex task. People have been using learning tools from diverse fields such as financial mathematics and machine learning to make trustworthy forecasting on such markets. However, the accuracy of such techniques had not been adequate. In order to forecast stocks accurately, it is crucial to understand the context-specific dependence of stock prices on their past values. Our approach offers superior accuracy as the training increases when it proceeds through the testing period through sliding window method.

PROBLEM STATEMENT

- Stock forecasting is a challenging task in the financial domain, as accurate predictions of stock prices are crucial for making informed investment decisions.
- Traditional methods often fall short in capturing the nonlinear patterns and dynamic nature of stock market data, leading to less reliable predictions.
- The objective is to create a robust and scalable forecasting solution that can handle large volumes of data, adapt to changing market conditions, and provide accurate predictions over different forecasting horizons.
- By addressing these challenges and developing a robust forecasting solution, this research aims to contribute to the field of financial forecasting and provide valuable tools for the financial industry.

EXISTING SYSTEM

Existing Algorithms :

- Autoregressive Integrated Moving Average (ARIMA)
- Long Short-Term Memory Networks (LSTM)
- Convolutional Neural Networks (CNN)
- Support Vector Machines (SVM)
- Random Forest (RF)
- Gaussian Processes (GPs)

Disadvantages :

- Data Dependency
- Limited Interpretability
- Overfitting and Generalization
- Computational Complexity
- Computational Complexity

PROPOSED SYSTEM

Proposed Algorithms :

- Recurrent Neural Networks (RNN)
- Long Short-Term Memory Networks (LSTM)
- Bi-directional Long Short-Term Memory Networks (Bi-LSTM)
- FBProphet

Advantages :

- Accurate Forecasting
- Automation and Speed
- Adapting to Changing Market Conditions
- Diverse Model Choices
- Reduced Human Bias

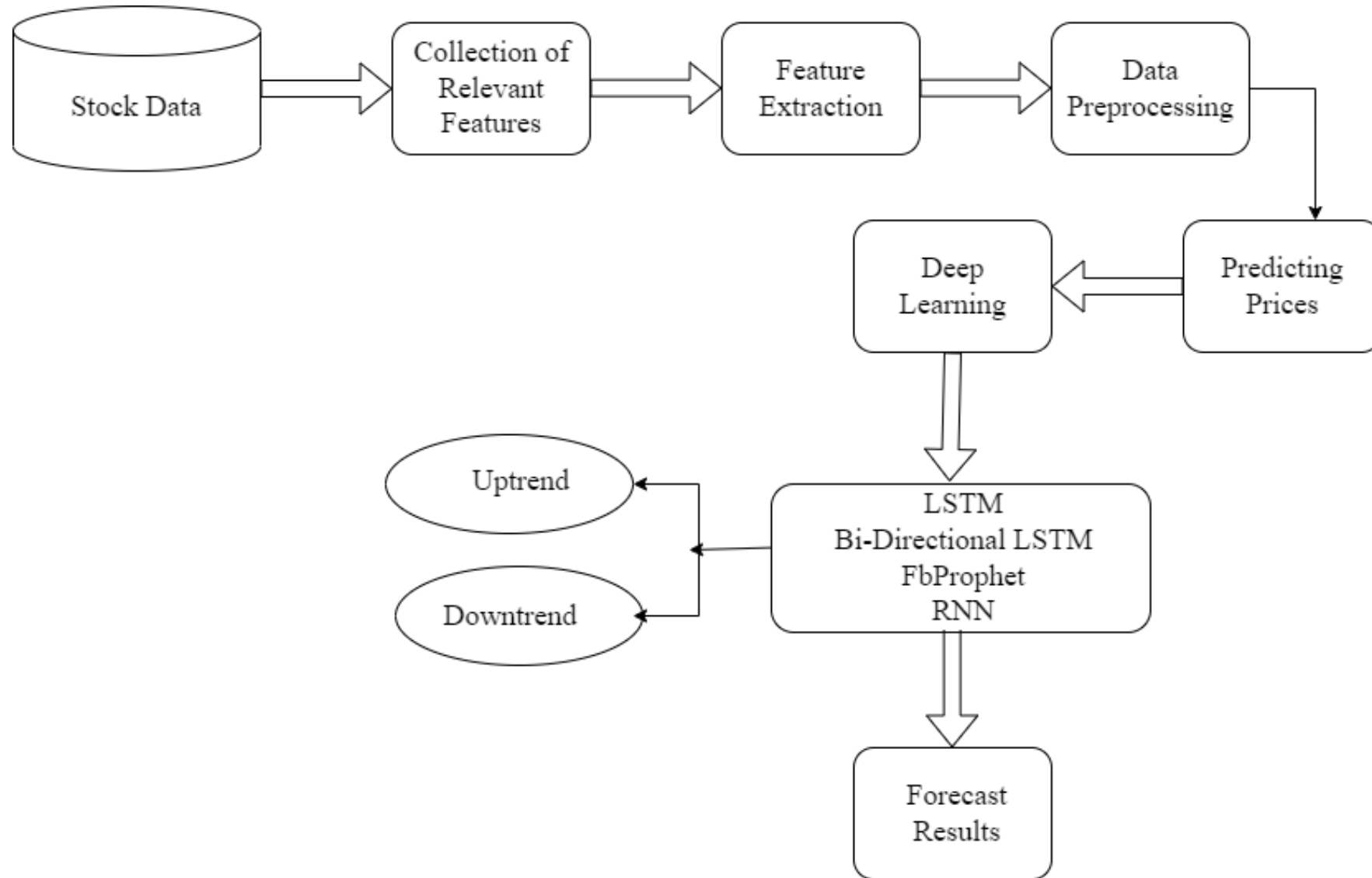
LITERATURE REVIEW

S.No	TITLE	ALGORITHM USED	ADVANTAGES	DISADVANTAGES
1	Novel optimization approach for stock price forecasting using multi-layered sequential LSTM	Multi-layered LSTM	Mean Error Percentage -2.18%	No use of dynamic data.
2	Multi-scale local cues and hierarchical attention-based LSTM for stock price trend prediction	Multi-scale local cues LSTM	Attention – Based model.	MLCA-LSTM degenerates more slowly than LSTM.
3	Real-time forecasting of time series in financial markets using sequentially trained dual-LSTMs	Autoregressive integrated moving average model	Dual – LSTM model.	Over-fitting issues

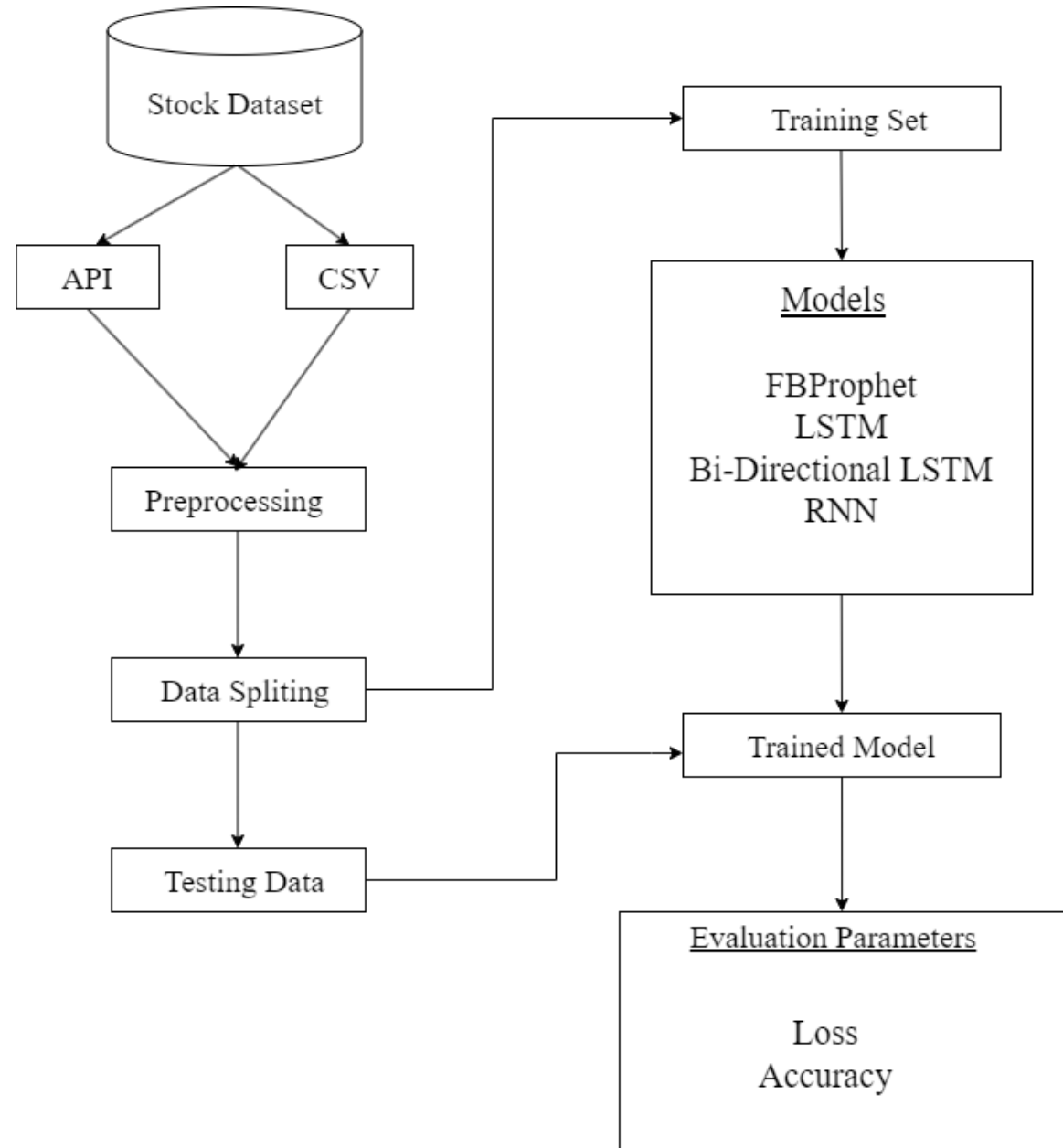
MODULE DESCRIPTION

- DATA COLLECTION
- DATA PREPROCESSING
- MODEL CREATION AND TRAINING
- EVALUATION
- DEPLOYMENT

PROPOSED SYSTEM



PROPOSED WORKFLOW

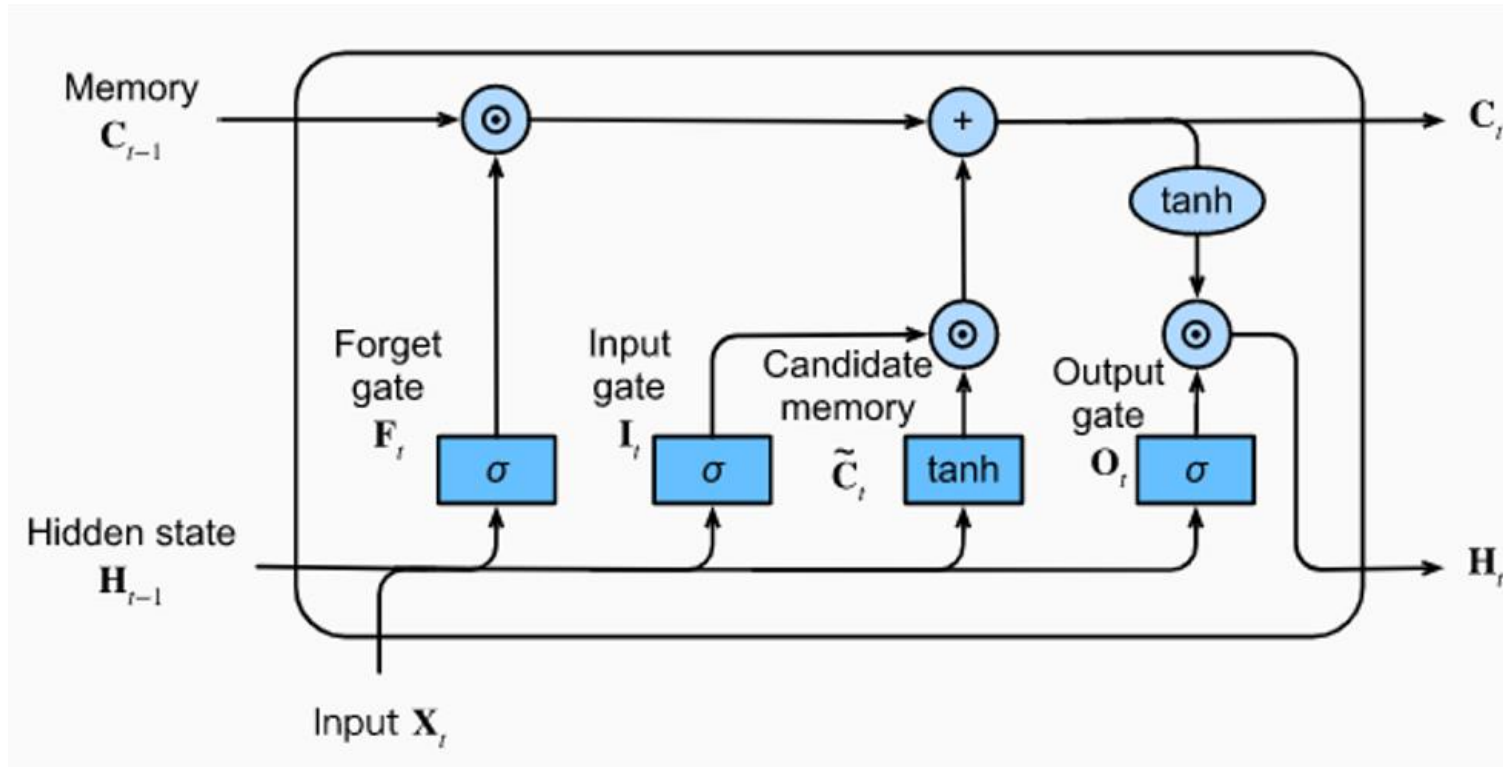


LONG SHORT-TERM MEMORY (LSTM)

Long Short-Term Memory Networks (LSTM), is a variety of recurrent neural networks (RNNs) that are capable of capturing long-term dependencies and process sequential information.

- **Forget Gate(f)** : The forget gate determines which information from the previous time step is irrelevant and should be discarded. This gate decides which values in the cell state to forget by assigning weights between 0 and 1 to each element.
- **Input Gate(i)** : The input gate determines which new information should be stored in the cell state. It takes the previous hidden state and the current input, passes them through a sigmoid activation function, and produces an input gate output.
- **Output Gate(o)** : The output gate determines the output of the LSTM cell based on the updated cell state. The output gate regulates the information flow from the cell state to the hidden state and generate the output.

LSTM ARCHITECTURE



$$f_t = \sigma_g (W_f \times x_t + U_f \times h_{t-1} + b_f)$$

$$i_t = \sigma_g (W_i \times x_t + U_i \times h_{t-1} + b_i)$$

$$o_t = \sigma_g (W_o \times x_t + U_o \times h_{t-1} + b_o)$$

$$c'_t = \sigma_c (W_c \times x_t + U_c \times h_{t-1} + b_c)$$

$$c_t = f_t \cdot c_{t-1} + i_t \cdot c'_t$$

$$h_t = o_t \cdot \sigma_c(c_t)$$

f_t is the forget gate

i_t is the input gate

o_t is the output gate

c_t is the cell state

h_t is the hidden state

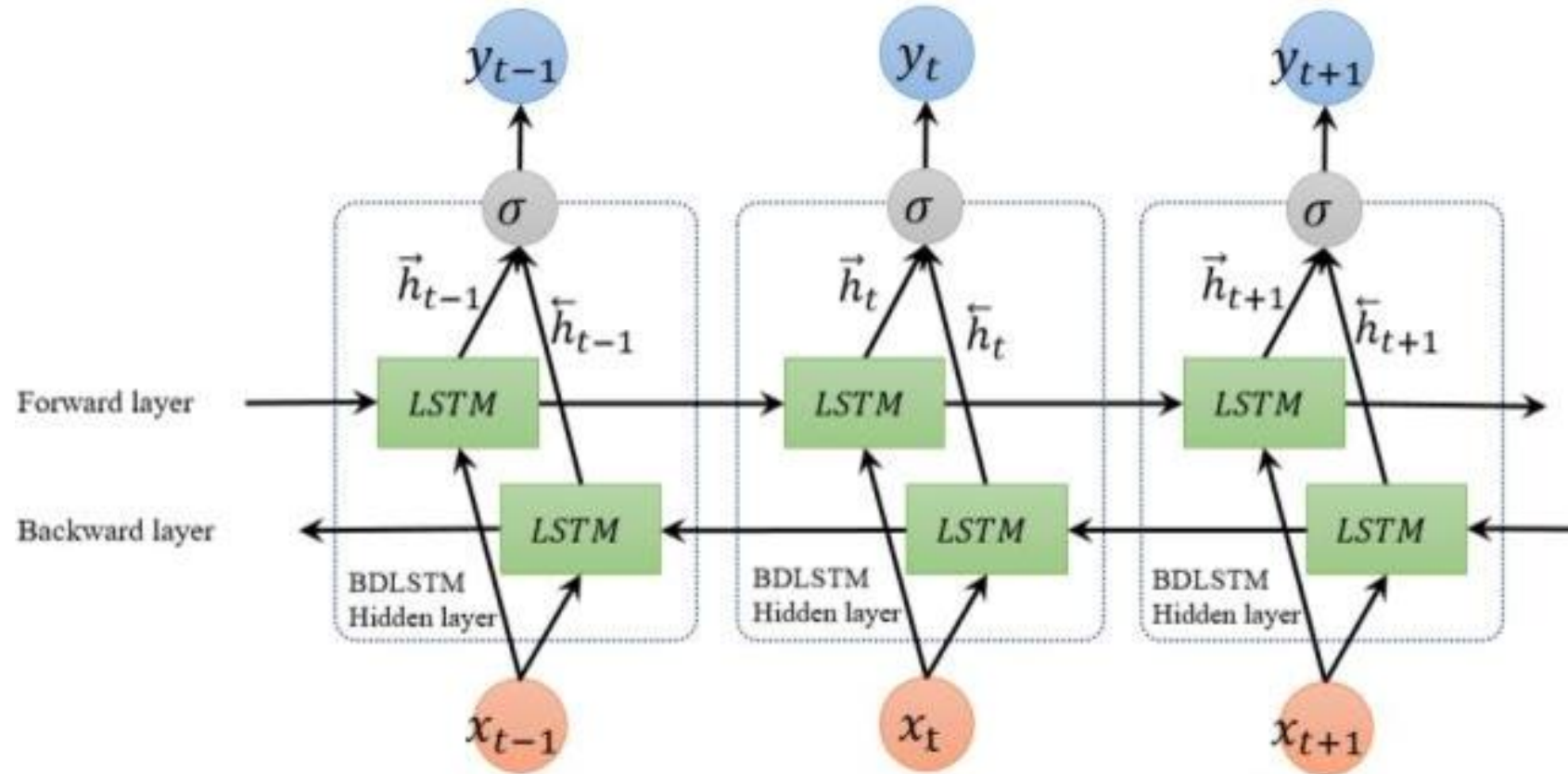
BI-DIRECTIONAL LONG SHORT-TERM MEMORY (BI-LSTM)

- In bidirectional LSTMs, each training sequence is presented forward and backward so as to separate recurrent nets. Both sequences are connected to the same output layer. Bidirectional LSTMs have complete information about every point in a given sequence, everything before and after it. Using bidirectional will run your inputs in two ways, one from past to future and one from future to past.

FEATURES :

- Bidirectional Processing
- Robust
- Attention Mechanisms
- Ensemble and Stacking

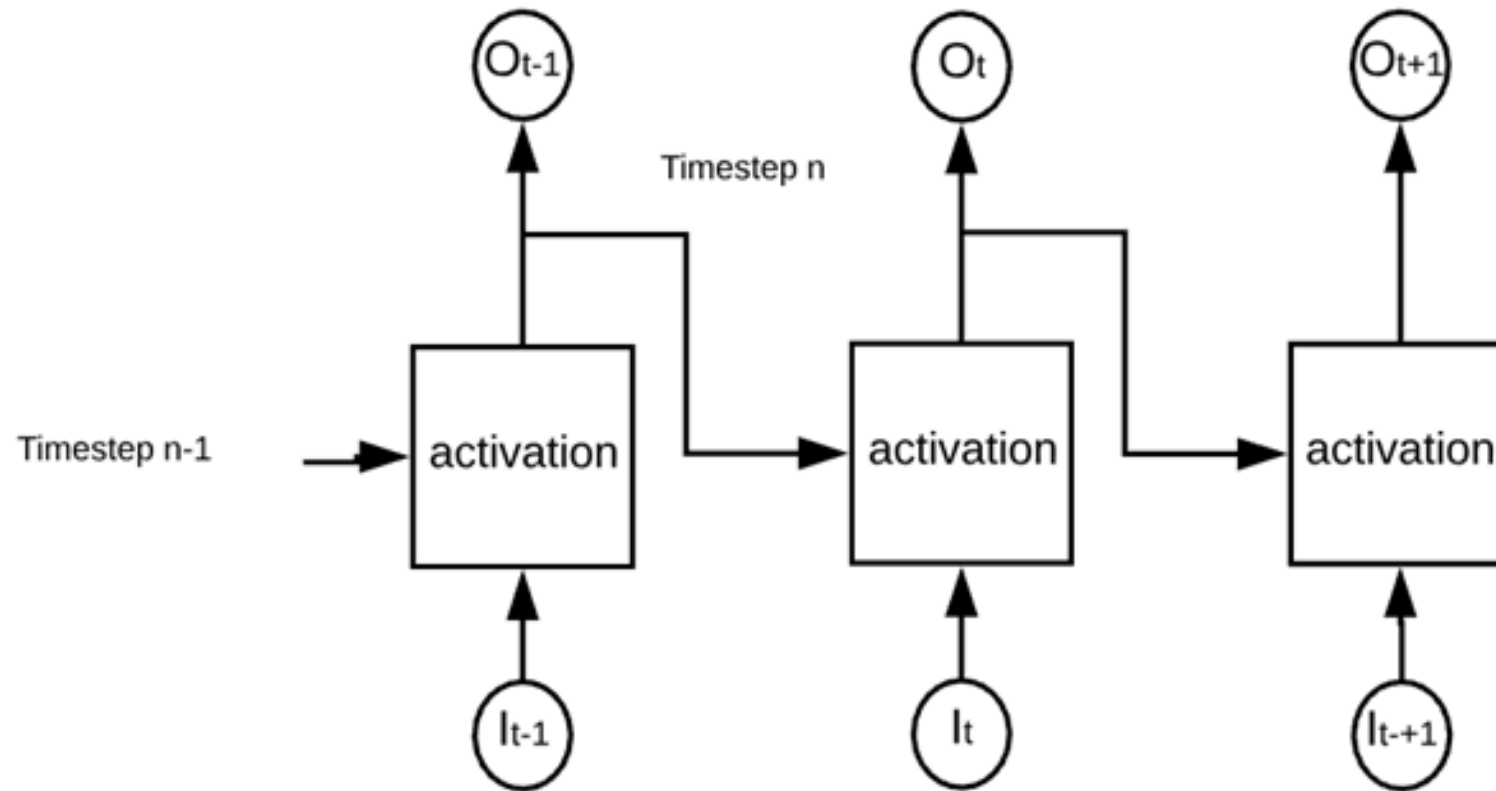
BI-LSTM ARCHITECTURE



RECURRENT NEURAL NETWORK (RNN)

- Recurrent Neural Network(RNN) is a type of Neural Network where the output from the previous step are fed as input to the current step. Recurrent Neural Networks (RNNs) are a class of neural networks designed for processing sequential data, where the order of elements in the sequence is significant. RNNs have a recurrent connection that allows them to maintain internal memory and capture temporal dependencies. RNN have a “memory” which remembers all information about what has been calculated.
- **FEATURES :**
 - Backpropagation
 - Maintaining Memory
 - Language Modeling
 - Bi-directional Processing

RNN ARCHITECTURE



FBPROPHET

- FBProphet is a forecasting library open-sourced by Facebook's Core Data Science team. It is based on an additive model where non-linear trends are fit with yearly, weekly, and daily seasonality, plus holiday effects. It allows for modeling of multiple seasonality, changing growth rates, and can automatically detect changes in trends by implementing a piecewise linear regression model.

FEATURES :

- User-friendly interface
- Highly Customizable
- Minimal Code
- Tunable Forecasts.
- Fully Automatic.

EFFICIENCY OF THE PROPOSED SYSTEM

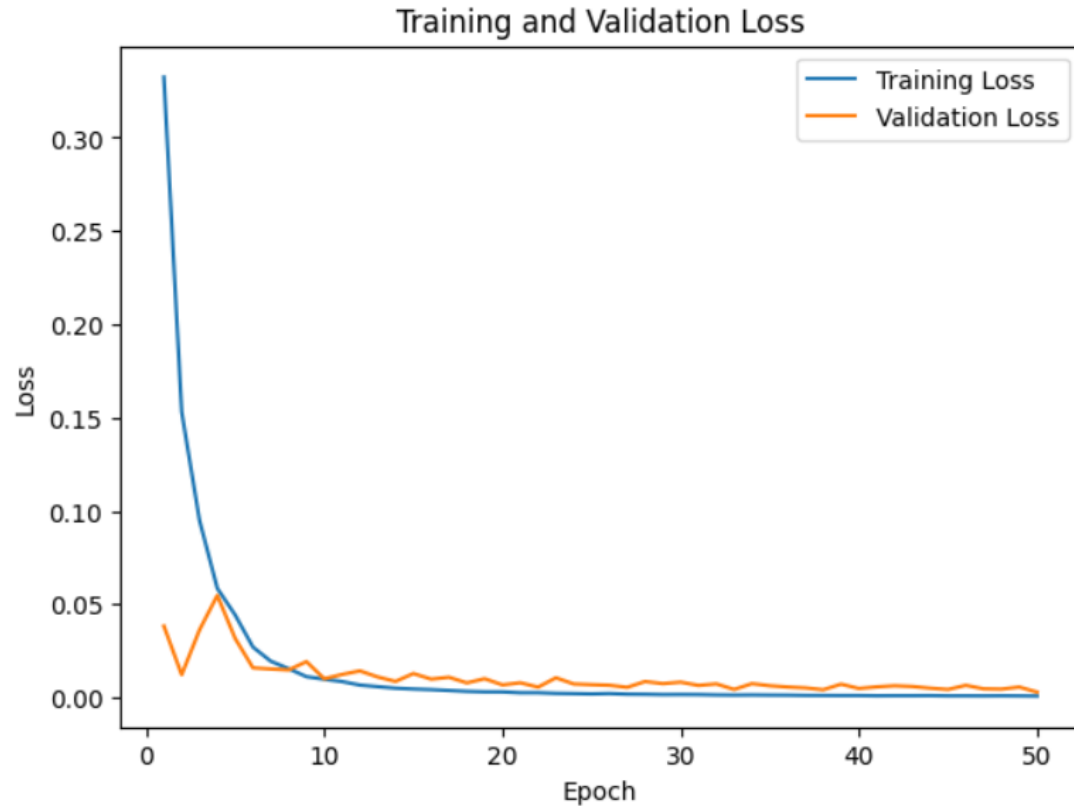
EXISTING SOLUTION

- DATASET - Static
- ALGORITHM USED :
 - Multi-layered Sequential LSTM – 2% Loss

PROPOSED SOLUTION

- DATASET - Dynamic
- ALGORITHM USED :
 - LSTM - 0.063 % Loss
 - BI-LSTM - 0.088 % Loss
 - RNN - 0.285 % Loss

LOSS RESULTS

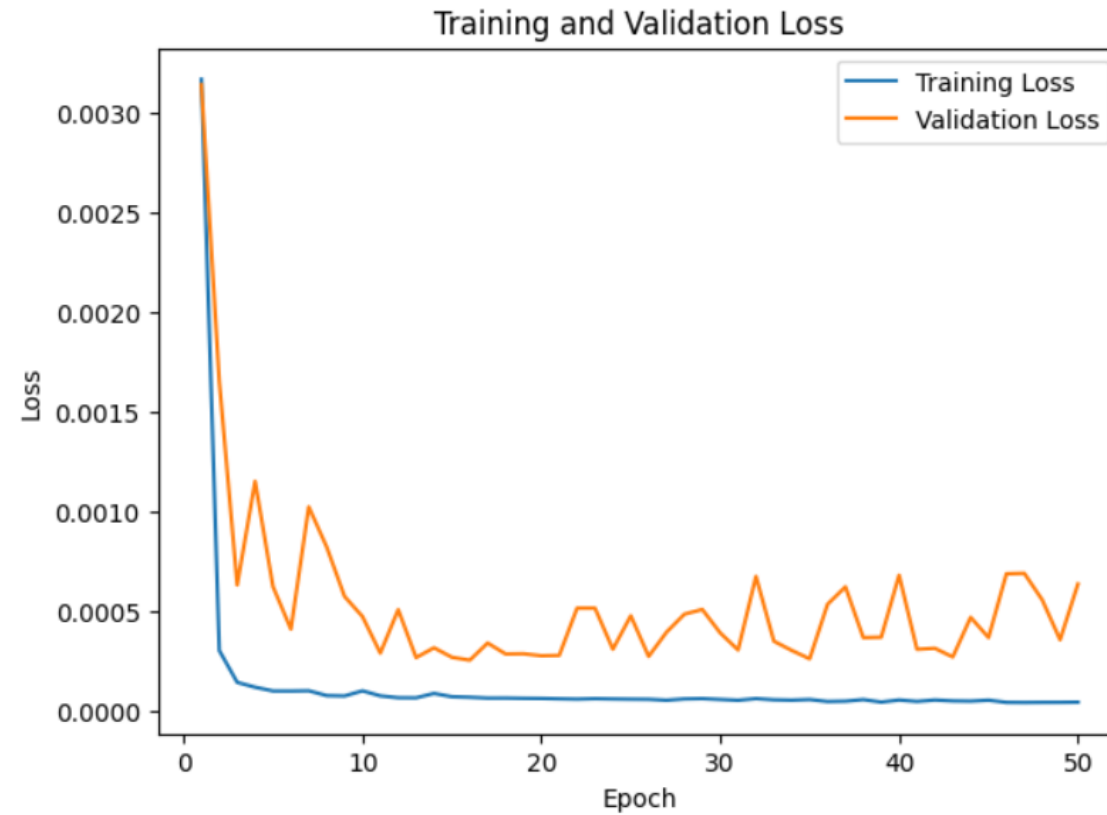


RNN MODEL



LSTM MODEL

LOSS RESULTS



BI-LSTM MODEL

MODEL METRICS

MODEL	EPOCHS	LOSS %
RNN	50	0.285
LSTM	50	0.063
BI-LSTM	50	0.088

HOME PAGE

Stock Price Prediction and Forecasting Using Machine Learning

- Stock Price Prediction using machine learning helps you discover the future value of company stock and other financial assets traded on an exchange.
- The entire idea of predicting stock prices is to gain significant profits. Predicting how the stock market will perform is a hard task to do.
- There are other factors involved in the prediction, such as physical and psychological factors, rational and irrational behavior, and so on.
- All these factors combine to make share prices dynamic and volatile.
- This makes it very difficult to predict stock prices with high accuracy.

PREDICTION PAGE

Stock Trend Forecasting

Select the algorithm

LSTM ▼

Enter Stock Ticker

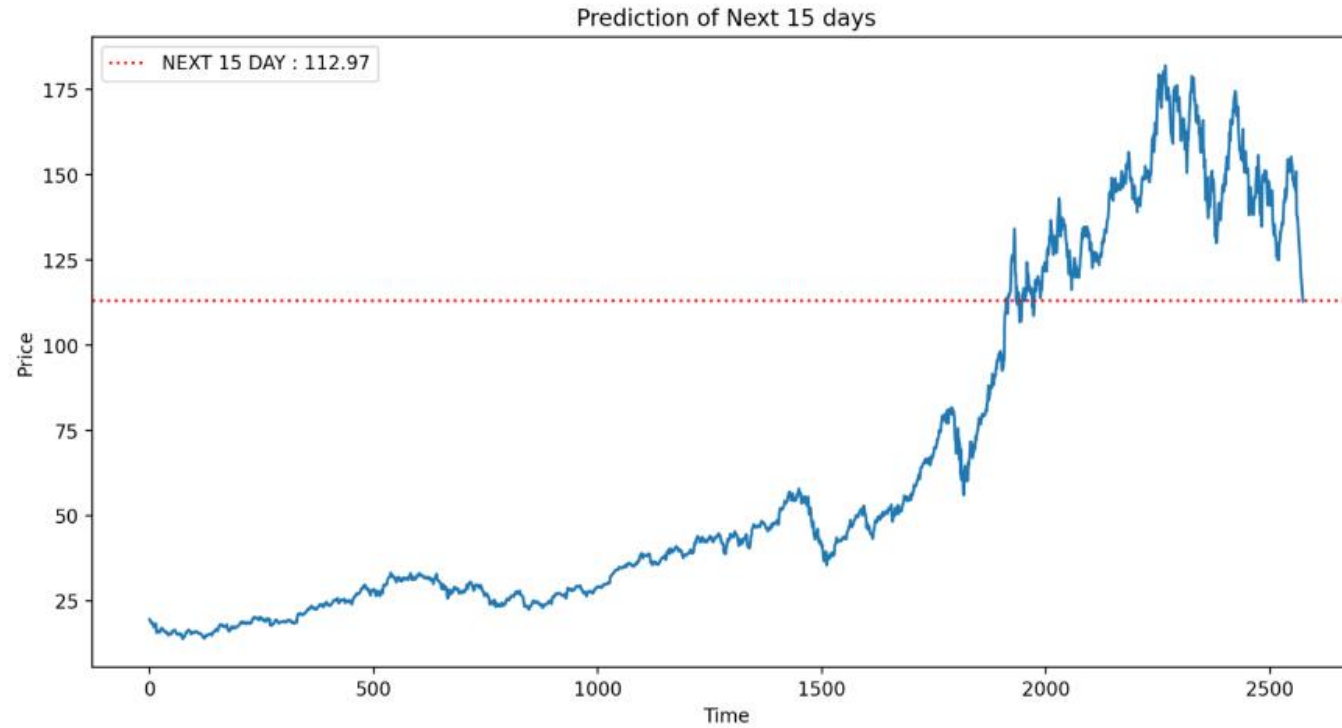
AAPL

Forecast Days

01545

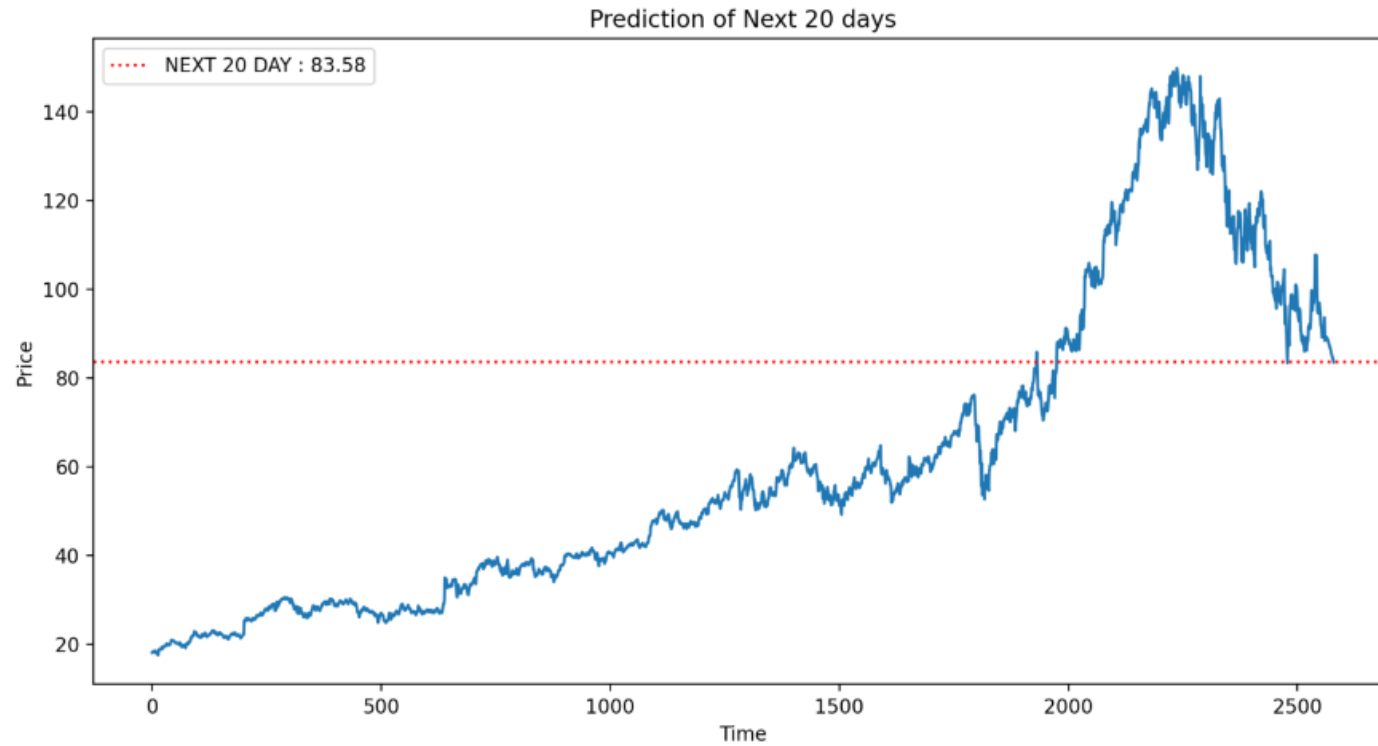
Submit

TEST CASE - 1



AAPL Stock Price Forecasting for next 15 days.

TEST CASE - 2



GOOGL Stock Price Forecasting for next 20 days.

REFERENCES

- [1] Novel Optimization Approach For Stock Price Forecasting Using Multi-Layered Sequential LSTM. Applied Soft Computing 134 (2023) 109830
- [2] Multi-scale local cues and hierarchical attention-based LSTM for stock price trend prediction. Neurocomputing 505 (2022) 92–100
- [3] Real-time forecasting of time series in financial markets using sequentially trained dual-LSTMs. Expert Systems With Applications 223 (2023) 119879
- [4] Forecasting Turning Points In Stock Price By Applying A Novel Hybrid CNN-LSTM-RESNET Model Fed By 2d Segmented Images. Engineering Applications of Artificial Intelligence 116 (2022) 105464
- [5] P. Mondal, L. Shit, S. Goswami, Study of effectiveness of time series modeling (ARIMA) in forecasting stock prices, Int. J. Comput. Sci. Eng. Appl. 4 (2) (2014) 13.
- [6] J. Bergstra, Y. Bengio, Random search for hyper-parameter optimization, J. Mach. Learn. Res. 13 (2) (2012).

REFERENCES

- [7] G.J. Deboeck, Trading on the edge: neural, genetic, and fuzzy systems for chaotic financial markets 39 (1994).
- [8] F. Feng, H. Chen, X. He, J. Ding, M. Sun, T.S. Chua, Enhancing stock movement prediction with adversarial training, IJCAI (2019)
- [9] K. Chen, Y. Zhou, F. Dai, A LSTM-based method for stock returns prediction: A case study of China stock market, in: 2015 IEEE International Conference on Big Data, big data, IEEE, 2015, pp. 2823–2824.
- [10] F. Jiang, G. Tang, G. Zhou, Firm characteristics and Chinese stocks, J. Manage. Sci. Eng. 3 (4) (2018) 259–283.
- [11] A. Yadav, C.K. Jha, A. Sharan, Optimizing LSTM for time series prediction in Indian stock market, Procedia Comput. Sci. 167 (2020) 2091–2100.
- [12] Y. Zhao, Y. Shen, Y. Zhu, J. Yao, Forecasting wavelet transformed time series

QUERIES ?

THANK YOU